

Units of concentration

Molarity and its scaled values

(Skoog: *equilibrium* Molarity)

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

$$\text{mM} = \frac{\text{mmol solute}}{\text{L solution}}$$

$$\mu\text{M} = \frac{\mu\text{mol solute}}{\text{L solution}}$$

Example:

Express 322 nM in mM

SOLUTION:

nM = nmoles/L and mmols = mmol/L

$$\begin{aligned} & 322(\text{nmoles/L}) * (1\text{mmol}/10^6\text{nmole}) \\ & = 3.22 \times 10^{-4} \text{mmol/L} \end{aligned}$$

Units of concentration

Equivalent units of molarity

$$M = \text{mol/L} = \text{mmol/mL}$$

$$\text{mM} = \text{mmol/L} = \mu\text{mol/mL}$$

$$\mu\text{M} = \mu\text{mol/L} = \text{nmol/mL}$$

Example: A solution was prepared by adding 600.0 μmoles of solute to 2.00 mL of solution. Express its concentration in M

SOLUTION:

$$(600.0\mu\text{mol}/2.00\text{mL})*(1\text{mmol}/10^3\mu\text{mol}) = 0.300\text{ M}$$

Units of concentration

Formality, F
(Skoog: analytical Molarity)

$F = (\text{Total moles solutes used to prepare a solution}) / \text{L solution}$

Non electrolyte $M = F$.

Weak electrolyte $M < F$.

Strong electrolyte $M = 0$ for the molecular form.

Differentiate between Formality, F (Skoog: analytical Molarity) and Molarity, M (Skoog: equilibrium molarity)

Example 4-4

Calculate Molarity and Formality of trichloroacetic acid in a solution prepared by dissolving 285 mg of the acid in 10 mL water (the acid is 73% ionized in water). (fw TCA = 163.4)

$$c_{HA} = (285 \text{ mg HA} / 10.00 \text{ mL solution}) * (1 \text{ mmol HA} / 163.4 \text{ mg HA}) = 0.174 \text{ mmol/mL} = 0.174 \text{ M}$$

$$[HA] = 0.27 * c_{HA} = 0.047 \text{ M}$$

Concentration expressed as p-function

$$\text{pX} = -\log[\text{X}]$$

Example:

pNa of a 2.00×10^{-3} M Na_2SO_4 solution

SOLUTION

$$\text{pNa} = -\log [\text{Na}^+],$$

$$[\text{Na}^+] = \text{Molarity of Na}^+ = 2 * 2.00 \times 10^{-3} \text{ M (formula stoichiometry)}$$

$$\text{pNa} = -\log(4.00 \times 10^{-3}) = 2.398$$

pH of a 5.4×10^{-4} M HCl solution

SOLUTION:

$$\text{pH} = -\log(5.4 \times 10^{-4}) = 3.27$$

Definition & equivalent units of w/v

For dilute aqueous solutions, density ~ 1
i.e. 1 g solution = 1 mL solutions.

$$\mu\text{g/mL} = \mu\text{g/g} = \text{w/w ppm} = \text{w/v ppm}$$

$$\text{w/v \%} = \text{g/100 mL} = \text{kg/100 L} = \text{g/dL}$$

$$\text{w/v ppm} = \text{g/10}^6 \text{ mL} = \mu\text{g/mL} = \text{mg/L}$$

$$\text{w/v ppb} = \text{g/10}^9 \text{ mL} = \text{ng/mL} = \mu\text{g/L}$$

$$1 \text{ cm}^3 = 1 \text{ cc} = 1 \text{ mL}$$

sometimes %, ppm, and ppb are used without displaying w/v when reporting concentration of solid solute in liquid solution. Example: NaCl in sea water, Ca in milk

Example: for a dilute aqueous solutions ($d = 1\text{g/mL}$)

$$\begin{aligned} 1\text{ppm(w/w)} &= (1 \mu\text{g solute/g solution}) \times (1 \text{ g solution/mL solution}) = (1 \mu\text{g solute/mL solution}) \\ &= 1\text{ppm (w/v)} \end{aligned}$$

Example 1

Express the concentration of 50% w/w NaOH in molarity ($d_{\text{NaOH}} = 1.54$ and $fw_{\text{NaOH}} = 40.00$)

SOLUTION:

$(50 \text{ g NaOH}/100 \text{ g solution}) * (1 \text{ mole NaOH}/40.00 \text{ g NaOH}) * (1.54 \text{ g solution}/1 \text{ mL}) * (1000 \text{ mL}/\text{L})$
 $\sim 19 \text{ mol/L} = 19\text{M}$

Example 2

How many milligrams of Na_2SO_4 are required to make 100 mL of _____

(1) 100 ppm (w/v) Na^+ solution?

SOLUTION:

$$(100 \mu\text{g Na}^+/\text{mL}) \times (100 \text{ mL}) \times (142.04 \mu\text{g Na}_2\text{SO}_4 / 2 \times 22.99 \mu\text{g Na}) \times (1 \text{ mg Na}_2\text{SO}_4 / 1000 \mu\text{g}) = 31 \text{ mg Na}_2\text{SO}_4$$

Example 3

A solid sample contains about 200ppm (w/w) of Ca.
How many mg of sample must be dissolved in 100mL in order to give a solution that is about 1ppm (w/v) after dilution?

SOLUTION:

$$1\mu\text{gCa/mL} * 100\text{mL} * 1\text{g solid}/200\mu\text{g Ca} * 100\text{ mg solid/g solid} = 500\text{ mg solid}$$

Example 4

It is recommended that drinking water contain 1.6 ppm fluoride (F^-) for prevention of tooth decay. A certain reservoir contained an estimated 1.59×10^{12} mL water.

How many kg of NaF should be added to give 1.6 ppm F^- ?
(fw NaF = 41.988 , fw F^- = 18.998)

SOLUTION:

$$(1.6 \mu\text{g } F^-/\text{mL solution}) * (1.59 \times 10^{12} \text{ mL}) * (41.988 \mu\text{g NaF} / 18.998 \mu\text{g } F^-) * (1 \text{ kg} / 10^9 \mu\text{g}) \\ = 5.6 \times 10^3 \text{ kg NaF}$$

Example 5

1.25 g of K_2SO_4 was dissolved in a 100 mL volumetric flask and diluted to volume. A 1.00 mL aliquot of the resulting solution was diluted to 1-L in a volumetric flask. What is the ppm of K?

(aw K = 39.098; fw K_2SO_4 = 174.254)

SOLUTION:

$$(1.25 \text{ g } \text{K}_2\text{SO}_4 / 100 \text{ mL A}) * (1.00 \text{ mL A} / 1000 \text{ mL B}) * (2 * 39.098 \text{ g K} / 174.254 \text{ g } \text{K}_2\text{SO}_4) * (10^6 \mu\text{g} / 1 \text{ g}) = 5.61 \mu\text{g K/mL B}$$

Example 6

A 13 g sample of wheat germ was dissolved in 50 mL of nitric acid and boiled nearly to dryness to remove most of the organic mass. The resulting solution was transferred to a 100 mL volumetric flask and diluted to volume. A 2 mL aliquot of this solution was diluted to 500 mL in a volumetric flask. The concentration was determined to be 5.76 ppm K. What is the w/w % K in the wheat germ?

SOLUTION:

$$(5.76 \mu\text{g K/mL B}) * (500 \text{ mL B} / 2 \text{ mL A}) * (100 \text{ mL A}) * (1 \text{ g K} / 10^6 \mu\text{g K}) / 13 \text{ g wheat germ}) * 100\% = 1.1 \text{ w/w\% K}$$

Example 7

An 89 ppm As standard was prepared from As_2O_3 by the following scheme. The initial sample was dissolved in 2.5 mL of NaOH and diluted to 100 mL in a volumetric flask. A 5 mL aliquot of the resulting solution was pipetted into a 1 L volumetric flask and diluted to volume. What weight in g of As_2O_3 was required?

(aw As = 74.9216; fw As_2O_3 = 197.8414)

Assigned to students to solve at home

Example 8

How many milliliters of 3.00 M H_2SO_4 are required to react with 4.35 g of solid containing 23.2% (w/w)

$\text{Ba}(\text{NO}_3)_2$.

The reaction is: $\text{Ba}^{++} + \text{SO}_4^{--} \rightarrow \text{BaSO}_4(\text{s})$

(fw $\text{Ba}(\text{NO}_3)_2 = 261$)

SOLUTION:

$4.35 \text{ g solid} * (23.2 \text{ g Ba}(\text{NO}_3)_2 / 100 \text{ g solid}) * (1 \text{ mole Ba}^{2+} / 261 \text{ g Ba}(\text{NO}_3)_2) * (1 \text{ mole SO}_4^{2-} / \text{mole Ba}^{2+}) * (1 \text{ mole H}_2\text{SO}_4 / \text{mole SO}_4^{2-}) * (1 \text{ L H}_2\text{SO}_4 / 3 \text{ mole H}_2\text{SO}_4) * (1000 \text{ mL/L}) = 1.29 \text{ mL H}_2\text{SO}_4$